[INCH-POUND]
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COMMERCIAL ITEM DESCRIPTION

SQUARES, MACHINISTS' COMBINATION

The General Services Administration has authorized the use of this commercial item description as a replacement of Federal Specification GGG-S-656E for Type VI, Class A, Styles A & B Machinists' Combination Squares for all federal agencies.

1. SCOPE. This commercial item description covers Styles 1 and 2, Machinists' combination center and miter head squares.

2. CLASSIFICATION. The machinists' combination square shall conform to the following styles:

Style 1: One side of the blade shall have divisions of 8ths of an inch on one edge and 16ths of an inch on the other edge; the opposite side of the blade shall have divisions of 32nds of an inch on one edge and 64ths of an inch on the other edge.

Style 2: One side of the blade shall have divisions of 32nds of an inch on one edge and 64ths of an inch on the other edge; the opposite side of the blade shall have divisions of 50ths of an inch on one edge and to 100ths of an inch on the other edge.

3. SALIENT CHARACTERISTICS.

3.1 Components. The square shall consist of a graduated blade, a square and miter head with a blade-clamping device, a level glass, and a scriber. A cast iron center head and protractor head shall be furnished if specified.

3.2 Materials. The component parts shall be fabricated from a fine grained, seasoned gray iron casting of uniform quality with a durable dark colored, smooth, or ripple finish on surfaces that are not machined or polished and free of detrimental defects. The blade shall be steel, properly hardened, tempered, and bright polished. The scriber shall be hardened steel. If a center head is specified, it shall be of cast iron. If a protractor head is specified, the body shall be of cast iron and of single(nonreversible) or double (reversible) type, as specified by the procuring agency. The dial of the protractor head shall be of cast iron, bright ground, chrome-plated satin finish,

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or a stainless steel plate, which is permanently secured to the cast iron dial and accurately fitted inside a circular space in the body. The level glasses shall be fabricated of substantial glass or plastic tubing properly. The liquid in the glass level shall be either clear or amber colored ("Cats eye"), at the option of the manufacturer. The liquid shall not discolor, cloud, or noticeably change in fluidity with age or at temperature as low as -20° Fahrenheit (F) and as high as 125°F.

3.3 Blade. The blade shall conform to the dimensions specified in Table 1. for length designated. The blade shall have a full-length groove along the center of one side to provide engagement with the clamping device of each head. The ruling edges and face shall be straight and parallel to each other. The blade graduations shall be permanently etched and read in reverse directions and the graduations and figures shall have a finish, which will make it possible for each accurate reading.

TABLE I. Dimensions, Style 1 & 2

	Blade				Square and Miter Head, Length	
	Width		Thickness			
Length (In.)	Min. (In.)	Max. (In.)	Min. (In.)	Max. (In.)	Min. (In.)	Max. (In.)
6	11/16	15/16	0.075	0.095	2-7/8	3-1/8
12	15/16	1-1/8	0.075	0.095	4	4-3/4
24	15/16	1-1/8	0.075	0.095	4-1/4	4-3/4

3.4 Graduations. The edges of graduation lines shall be sharp, straight, legible, and permanent, and shall extend to the reading edge. Blade graduations shall conform to Style 1 or Style 2 as specified. The graduations shall be a uniform width and depth. The maximum variation in width between like types of graduations and in the individual graduation shall be such that the maximum width shall not exceed the minimum width by more than 35 percent of the minimum width. The corners of each graduated edge shall be marked to show the smallest fractional subdivision to which it is graduated. The subdivisions shall be marked with proper figures at the corresponding inches.

3.4.1 Style 1. One side of the blade shall have divisions of 8ths of an inch on one edge and 16ths of an inch on the other edge; the opposite side of the blade shall have divisions of 32nds of an inch on one edge and 64ths of an inch on the other edge.

3.4.2 Style 2: One side of the blade shall have divisions of 32nds of an inch on one edge and 64ths of an inch on the other edge; the opposite side of the blade shall have divisions of 50ths of an inch on one edge and to 100ths of an inch on the other edge.

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3.5 Square and Miter Head. The square, miter, and horizontal surfaces, including their adjacent side surfaces, shall be accurately finished. There shall be a significant slot for insertion of the blade. The clamping device shall have a substantial knurled thumbscrew or thumbnut suitable for rigidly clamping the head of any point on the blade. The locking device shall be so designed to prevent slipping from the head when the blade has been removed. The level glass to be parallel with the square face of the head, and the scriber shall be pointed, provided with a knurled head, and be either mechanically or frictionally held in a brass bushed hole in the head.

3.6 Center Head. When specified, a center head shall be furnished. The inside area surfaces, including their adjacent side surface, shall be accurately finished straight, true and square. The arms shall include a 90° angle with a tolerance of ±15 minutes with the vertex of the arms centering the edge of the blade. Either arm of the center head shall be at 1/2 the included angle ±3 minutes with either edge of the blade. There shall be a significant slot for insertion of the blade and a substantial knurled thumbscrew or thumbnut suitable for securely holding the head at any point on the blade. The locking device shall be so designed to prevent slipping from the head when the blade has been removed. The arm length, measured along the inside surface, on 6 inch blade squares shall be 2 inches ±1/8 inch. On squares with 12 inch or longer blades, the arms shall be 2 3/8 inches ±1/8 inch.

3.7 Protractor Head. When specified, a protractor head shall be furnished. When used in conjunction with the graduated blade, it shall be suitable for use as a bevel protractor. The protractor head shall consist of a body, a turret-type dial with level glass, and 2 or more clamping devices with substantial knurled thumbscrews or thumbnuts. The locking device shall be so designed to prevent slipping from the head when the blade has been removed. One clamping device shall be suitable for rigidly clamping the protractor body to the blade and the other shall be suitable for rigidly clamping body to the dial.

3.7.1 Protractor Body. The straight edges and finished portion of the protractor body shall be finished straight, true and square and shall be not less than 6 3/8 inches nor more than 7 1/4 inches long. The body shall have 1 or more zero index lines, each marked with a figure zero. The single (non-reversible) type of body, as specified, shall have a single straight edge arranged to extend over one face of the graduated blade and capable of complete rotation about the dial. The double (reversible) type of body, as specified, shall have 2 equal length straight edges, each extending equal distance over opposite faces of the graduated blade. One straight edge of the double (reversible) type body may be detachable, at the option of the user.

3.7.2 Dial. The dial shall be accurately fitted inside a circular space in the body. The peripheral surface of the dial and the inside surface of the circular space of the body shall be accurately finished. The dial shall be graduated over an arc of 180°, and graduations and figures shall be arranged to read from zero to 90° in either direction at the zero index, or from zero to 180° with two sets of figures reading in opposite directions. The angle graduations shall be subdivided to 1° and marked to show each 10°. The spirit level shall be permanently secured to the dial and arranged parallel to the straight edges of the blades. The graduations shall have durable filling of black color.

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3.8 Level glasses. The level glasses shall have permanent and readily distinguishable black bubble defining lines, suitable for properly centering the bubble. Level shall be permanently secured to the head and enclosed by a metal housing in a manner such that only the portion necessary for observing the bubble positions is exposed.

3.9 Tolerances and Accuracy.

3.9.1 Blade Length Tolerance. The blade length tolerance for 6 inch blade shall be -0.002 to +0.004 inches. The blade length tolerance for 12 inch blade shall be -0.020 to +0.005 inches. The blade length tolerance for 24 inch blade shall be -0.012 to +0.007 inches. Both blade ends shall be square to both blade edges within an angle of 90° ±5 minutes.

3.9.2 Scale Centering Error. The scale centering error shall be measured from the leading edge of the first 1/2 inch graduation to the zero blade end and from the leading edge of the last 1/2 inch graduation to the opposite blade end. The scale centering error shall not exceed ±0.0025 inches for each measurement.

3.9.3 Scale Graduation Error. The scale graduation error shall conform to the specifications in Table 2.

Table 2. Scale Graduation Error

Length (In.)	Measurement	Error (In.)
First 6	Between left scale end and any graduation	- 0.0035 to + 0.004
$6 < X \leq 12$	Between left scale end and any graduation	- 0.0035 to + 0.005
$12 < X \leq 18$	Between the zero scale end and any graduation	- 0.0035 to + 0.006
$18 < X \leq 24$	Between the zero scale end and any graduation	- 0.0035 to + 0.007

3.9.4 Squareness of blade. The squareness of the blade to the surface of the square head shall be 90° ±3 minutes.

3.9.5 Miter. The out-of-miter error of the head with either edge of the blade shall not exceed 0.004 inch per foot.

3.9.6 Protractor. The error in any angular measurement on the scale with either graduated blade edge and either straight edge of the protractor body shall not exceed 15 minutes.

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3.9.7 Center Head. The blade shall bisect the included angle, as measured, of the center head within a tolerance of ±3 minutes. The level vial shall have a vial sensitivity not to exceed 50 minutes and shall be set in the square head parallel to the base within 0.012 inch per foot.

3.10 Marking. Mark each item in a permanent and legible manner with manufacturer's part number, manufacturer's name or identifying symbol, and state and country of manufacture. All markings shall be made directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item.

4. REGULATORY REQUIREMENTS.

4.1 In accordance with the section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

4.2 Unless otherwise specified, standard commercial tolerances shall apply to all dimensions.

5. PRODUCT CONFORMANCE. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

6. MARKET ACCEPTABILITY. The following market acceptability criteria apply.

6.2.1 The manufacturer producing this item must have been producing a product meeting the requirements of this CID for at least 2 years.

6.2.2 The manufacturer must have sold 1000 units meeting this CID in the commercial market place over the past 2 years.

7. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

8. NOTES.

8.1 Attachments shall be as specified in the acquisition document.

8.1.1 Graduations (see 3.4)

8.1.2 Center head (see 3.6)

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8.1.3 Protractor head* (see 3.7) (*Protractor heads are not available commercially for the 6 and 9 inch squares.)

8.1.4 Protractor body type (reversible or nonreversible) (see 3.7.1)

8.2 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	Style	Size
5210-00-224-9937	1	6 in.
5210-00-273-1937	1	12 in.
5210-00-450-6920	1	12 in.
5210-00-540-3513	1	12 in.
5210-00-078-8944	2	6 in.
5210-00-078-8947	2	12 in.
5210-00-078-8948	2	12 in.
5210-00-078-8949	2	12 in.
5210-00-078-8950	2	24 in.

MILITARY INTERESTS:

None: DoD has no registered interest in revisions and amendments to this Commercial Item Description until further notice.

PREPARING ACTIVITY:

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